

Kartik Upadhyay

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SUMMARY

Computer Science Engineering undergraduate with hands-on experience designing Retrieval-Augmented Generation (RAG) pipelines, LLM-based agentic workflows, and scalable RESTful backends. Combines deep foundations in Data Structures & Algorithms and Python (FastAPI) with production full-stack engineering (React, Next.js, Node.js) and automated testing (Cypress). Proven track record of building reliable, end-to-end AI applications with measured latency, structured schemas, and zero-downtime architecture.

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Vector Databases (ChromaDB, Pinecone, FAISS), Prompt Engineering, Pydantic, OpenAI & Gemini APIs, Hugging Face
Languages: Python, JavaScript (ES6+), Java, C++, SQL
Backend & Systems: FastAPI, Node.js, Express.js, RESTful API Design, Vector Search, Redis (Caching basics)
Frontend: React.js, Next.js (App Router), Tailwind CSS, Redux, HTML5, CSS3
Testing & Tools: Cypress (Automated E2E Testing), Git, GitHub, Docker, Postman, CI/CD Basics, SDLC
CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Complexity Analysis

EXPERIENCE

AI Operations & Growth Intern

Aug 2026 – Present

Kafalmart (Nyaari)

Remote

- Engineered automated data extraction and catalog workflows in Python using LLM function calling and Pydantic schemas, reducing manual catalog processing time by 65%.
- Implemented prompt engineering templates with structured JSON validation, minimizing model hallucinations and standardizing product descriptions across 500+ SKUs.
- Analyzed sales, operational metrics, and user retention; built automated weekly executive dashboards to inform founder-led growth strategies.
- Built lightweight API automations for customer FAQ triage, decreasing first-response turnaround for recurring inquiries.

Full Stack Developer Intern

Apr 2026 – Jul 2026

TechXAlt

Dehradun, Uttarakhand

- Architected and deployed 12+ production RESTful API endpoints in Node.js and Express powering core application features and high-volume transaction flows.
- Connected backend microservices with dynamic React.js frontends, ensuring responsive state synchronization and UI reliability.
- Authored automated end-to-end test suites using Cypress, catching critical regressions early and maintaining 90%+ API reliability prior to release.
- Participated in Agile sprint planning, peer code reviews on GitHub, and iterative delivery to ship stable weekly production updates.

PROJECTS

Portfolio AI Assistant & RAG Engine | *Next.js, React, RAG, Web Streams API, Tailwind CSS*

2026

- Engineered an end-to-end contextual RAG assistant embedded into a live portfolio, enabling recruiters to query technical background, projects, and stack interactively.
- Implemented keyword/semantic knowledge chunking and relevance scoring to retrieve grounded facts before inference, mitigating model hallucinations.
- Architected a dual-engine streaming pipeline supporting cloud LLMs (Gemini / OpenAI) with an intelligent zero-downtime local fallback, delivering word-by-word streaming via Web Streams API.

Multi-Step Loan Application System | *Node.js, Express, React, Cypress, RESTful APIs, OOP*

Dec 2025 – Feb 2026

- Developed a full-stack loan origination platform featuring dynamic multi-step form workflows, document uploads, digital signatures, and dynamic EMI calculation.

- Designed modular RESTful APIs with PAN and Aadhaar identity verification logic, applying clean object-oriented architecture and layered service patterns.
- Integrated client/server input validation, form auto-saving, and comprehensive Cypress E2E test coverage, achieving high system reliability and zero release regressions.

Autonomous Multi-Tool Agent & Copilot | *Python, FastAPI, LangChain, ChromaDB* 2026

- Developed an autonomous agent pipeline capable of multi-step planning, web search execution, and dynamic external tool calling to synthesize comprehensive technical briefings.
- Implemented vector store retrieval (ChromaDB) with recursive text chunking and similarity thresholding to deliver cited sources for generated summaries.
- Exposed inference endpoints via a high-performance FastAPI microservice with asynchronous request handling and Pydantic validation.

EDUCATION

Birla Institute of Applied Sciences	Bhimtal, Uttarakhand
<i>Bachelor of Technology in Computer Science and Engineering – CGPA: 7.64</i>	<i>Expected 2027</i>
• Senior Secondary (12th): 77% Secondary (10th): 84%	

CERTIFICATIONS & LEADERSHIP

Certifications: Applied Data Science Lab (WorldQuant University, 2026), AI Training (AcmeGrade, 2024), Geo-Data Sharing & Cyber Security (IIRS, ISRO Grade A+, 2024), Data Visualization & Frontend Libraries (freeCodeCamp)

Leadership: Elected Sports Secretary (directed inter-college tournaments for 500+ participants); Student Representative on the institute's AICTE Induction Cell (mentored incoming freshmen)